

S METFORMIN 850MG TABLET

Each tablet contains:- Metformin hydrochloride 850 mg
Calcium Carbonate
Croscarmellose
Lactose
Deionised Water
Talcum
Titanium Dioxide

Hydroxypropyl Methylcellulose
Isopropyl Alcohol
Polyethylene Glycol 6000 (PEG 6000)
Magnesium Stearate
Fusidate K/90 (PVP K-90)

Product Description:

Oral, convex, white, film-coated tablet with 'TD' debossed on one side and single score on the other.

Pharmacodynamics:

Metformin hydrochloride is a biguanide with antihyperglycaemic effects, on both basal and postprandial hyperglycaemia. It does not stimulate insulin secretion and therefore does not cause hypoglycaemia. Metformin reduces basal hyperinsulinaemia, and in combination with insulin, reduces insulin requirement.

Metformin exerts its antihyperglycaemic effect via multiple mechanisms:

Metformin reduces hepatic glucose production.

Metformin facilitates peripheral glucose uptake and utilization, in part by increasing insulin action. Metformin alters glucose turnover in the gut: Uptake from circulation is increased and absorption from food is decreased. Additional mechanisms attributed to the gut include an increase in release of glucagon-like peptide 1 (GLP-1) and a decrease of bile acid reabsorption. Metformin alters the gut microbiome.

Metformin can improve the lipid profile in hyperlipidaemic individuals.

In clinical studies, use of metformin was associated with either a stable body weight or modest weight loss.

Metformin is an adenosine monophosphate-activated protein kinase (AMPK) activator and increases the transport capacity of all types of membrane glucose transporters (GLUTs).

Pharmacokinetics:

Absorption

After an oral dose of metformin hydrochloride tablet, maximum plasma concentration (C_{max}) is reached in approximately 2.5 hours (t_{max}). Absolute bioavailability of a 500 mg or 850 mg metformin hydrochloride tablet is approximately 50-60% in healthy subjects. After an oral dose, the non-absorbed fraction recovered in faeces was 29-30%.

After oral administration, metformin hydrochloride absorption is saturable and non-linear. It is assumed that the pharmacokinetics of metformin hydrochloride absorption is non-linear.

At the recommended metformin hydrochloride doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours and are generally less than 1 microgram/ml.

In controlled clinical trials, maximum metformin hydrochloride plasma levels (C_{max}) did not exceed 5 microgram/ml, even at maximum doses.

Food decreases the extent and slightly delays the absorption of metformin hydrochloride. Following oral administration of a dose of 850 mg tablet, a 40% lower plasma peak concentration, a 25% decrease in AUC (area under the curve) and a 35 minute prolongation of the time to peak plasma concentration were observed. The clinical relevance of these findings is unknown.

Distribution

Plasma protein binding is negligible. Metformin hydrochloride partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean volume of distribution (V_d) ranged between 63-278 l.

Excretion

Metformin hydrochloride is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Renal clearance of metformin hydrochloride is ~400 ml/min, indicating that metformin hydrochloride is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 5.5 hours.

When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin hydrochloride in plasma.

Indications:

It is used in the treatment of maturity on-set diabetes mellitus not responsive to diet alone or diet plus treatment with a sulphonylurea.

Recommended Dosage:

Adults with normal renal function (GFRs >90ml/min)

Monotherapy and Combination with Oral Antidiabetic Agents

The usual starting dose is one tablet 2 or 3 times daily given during or after meals. After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve glycaemic control and tolerability.

The maximum recommended dose of metformin hydrochloride is 3 g daily, taken as 3 divided doses.

If transfer from another oral antidiabetic agent is intended, discontinue the other agent and initiate metformin at the dose indicated above.

Combination with Insulin

Metformin hydrochloride and insulin may be used in combination therapy to achieve better blood glucose control. Metformin hydrochloride is given at the usual starting dose of one tablet 2-3 times daily, while insulin dosage is adjusted on the basis of blood glucose measurements.

Older

Due to the potential for decreased renal function in elderly subjects, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary.

Renal Impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR (ml/min)	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional Considerations
45 - 60	3000 mg	Dose reduction may be considered in relation to declining renal function.
45 - 59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30 - 44	1000 mg	Metformin is contraindicated.
<30	-	-

Elderly population, Monotherapy and combination with insulin

850 Metformin (850mg) Tablet can be used in children from 10 years of age and adolescents. The usual starting dose is 500 mg or 850 mg metformin hydrochloride once daily, given during or after meals. After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose of metformin hydrochloride is 2 g daily, taken as 2 or 3 divided doses.

Route of Administration: Oral

Contraindications:

- Hypersensitivity to metformin or to any of the excipients.
- Any type of acute metabolic disorder (such as lactic acidosis, diabetic ketoacidosis).
- Diabetic pre-coma.
- Severely reduced kidney function (GFR <30ml/min) Acute conditions with the potential to alter renal function such as: dehydration, severe infection, shock.
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as: decompensated heart failure, respiratory failure, recent myocardial infarction, shock.
- Hepatic insufficiency, acute alcohol intoxication, alcoholism.

Warnings and Precautions:

Lactic Acidosis

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiovascular illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis. In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis (see Contraindications and Interaction with other medicinal products and other forms of interaction).

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by metabolic acidosis, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal Function

GFR should be assessed before treatment initiation and regularly thereafter, see Section Recommended Dosage. Metformin is contraindicated in patients with GFR<30 ml/min and should be temporarily discontinued in the presence of conditions that alter renal function, see Contraindications.

Cardiac Function

Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, metformin may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, metformin is contraindicated (see section Contraindications).

Administration of Individual Contraceptive Agents

Intravascular administration of individual contrast agents may lead to contrast induced nephropathy resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Surgery

Metformin must be discontinued at the time of surgery under general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Paediatric Population

The diagnosis of type 2 diabetes mellitus should be confirmed before treatment with metformin is initiated. No effect of metformin on growth and puberty has been detected during controlled clinical studies of one-year duration but no long-term data on these specific points are available. Therefore, a careful follow-up of the effect of metformin on these parameters in metformin-treated children, especially prepubescent children, is recommended.

Children Aged Between 10 and 17 years

Particular caution is recommended when prescribing to children aged between 10 and 17 years.

Other Precautions

All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.

The usual laboratory tests for diabetes monitoring should be performed regularly.

Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anaemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contra-indicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Metformin hydrochloride alone does not cause hypoglycaemia, but caution is advised when it is used in combination with insulin or other oral antidiabetics (e.g. sulphonylureas or meglitinides).

Interaction with Other Medicaments:

Concomitant use not recommended

Alcohol

Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in case of fasting, malnutrition or hepatic impairment.

Individualised contrast agents

Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Medicinal products with intrinsic hyperglycaemic activity (e.g. glucocorticoids (systemic and local routes) and sympathomimetics)

More frequent blood glucose monitoring may be required, especially at the beginning of treatment. If necessary, adjust the metformin dosage during therapy with the respective medicinal product and upon its discontinuation.

Organic cationic transporters (OCT)

Metformin is a substrate of both transporters OCT1 and OCT2.

Co-administration of metformin with:

• Inhibitors of OCT1 (such as ranitidine) may reduce efficacy of metformin.

• Inhibitors of OCT2 (such as cimetidine, disulfiram, resmetimide, trimethoprim, vandetanib, vandetanib, vandetanib) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.

• Inhibitors of both OCT1 and OCT2 (such as cotinine, elagatril) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/reducers may alter the efficacy of metformin.

Pregnancy and Lactation:

Discontinuation of hyperglycaemia in the periconceptional phase and during pregnancy is associated with increased risk of congenital abnormalities, pregnancy loss, pregnancy-induced hypertension, preeclampsia, and perinatal mortality. It is important to maintain blood glucose levels as close to normal as possible throughout pregnancy, to reduce the risk of adverse hyperglycaemia-related outcomes to the mother and her child.

Metformin crosses the placenta with levels that can be as high as maternal concentrations.

If clinically needed, the use of metformin can be considered during pregnancy and in the periconceptional phase as an addition or an alternative to insulin.

Breastfeeding

Metformin hydrochloride is excreted into human breast milk. No adverse effects were observed in breastfed newborns / infants. However, as only limited data are available, breastfeeding is not recommended during metformin hydrochloride treatment. A decision on whether to discontinue breastfeeding should be made, taking into account the benefit of breast feeding and the potential risk to adverse effects on the child.

Side Effects:

During treatment initiation, the most common adverse reactions are nausea, vomiting, diarrhoea, abdominal pain and loss of appetite which resolve spontaneously in most cases. To prevent them, it is recommended to take metformin in 2 or 3 daily doses and to increase slowly the doses.

The following adverse reactions may occur under treatment with metformin. Frequencies are defined as follows: very common >1/10; common >1/100; <1/100; uncommon >1/1,000; <1/1,000; rare >1/10,000; <1/10,000; very rare <1/100,000.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Metabolism and nutrition disorders

Common: Vitamin B12 decrease/deficiency (see Warnings and Precautions).

Very rare: Lactic acidosis (see Warnings and Precautions).

Nervous system disorders

Common: Taste disturbances.

Gastrointestinal disorders

Very common: Gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. To prevent them, it is recommended that metformin be taken in 2 or 3 daily doses during or after meals. A slow increase of the dose may also improve gastrointestinal tolerability.

Unstable laboratory results

Very rare: Isolated reports of liver function tests abnormalities or hepatitis resolving upon metformine discontinuation.

Skin and subcutaneous tissue disorders

Very rare: Skin reactions such as erythema, pruritus, urticaria.

Respiratory disorders

Adverse event reporting was similar in nature and severity to that reported in adults.

Symptoms and Treatment of Overdose:

Hyperglycaemia has not been seen with metformin hydrochloride doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose of metformin hydrochloride or concomitant risks may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and metformin hydrochloride is haemodialysis.

Effects on Ability to Drive and Use Machines:

Metformin hydrochloride monotherapy does not cause hypoglycaemia and therefore has no effect on the ability to drive or to use machines. However, patients should be alerted to the risk of hypoglycaemia when metformin hydrochloride is used in combination with other antidiabetic agents (sulphonylureas, insulin, repaglinide).

Storage Conditions:

Store at or below 30°C.

Malaysia
Pack Size: Bottle Pack: A bottle of 100 tablets.
Blister pack: 10 x 10, 50 x 10, 10 x 10 tablets per strip.

Singapore
Pack Size: Blister pack: 100 x 10, 50 x 10, 10 x 10 tablets per strip.

Pack Size (export only): 500 tablets.

Malaysia
Shelf Life: 3 Years

Singapore
Shelf Life: 3 Years

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.			
Manufacturer & Product Registration Holder:	Imported by:		
Seward Pharmaceutical Sdn. Bhd.	Seward Pharmaceutical Pte. Ltd.		
No. 9, 11 & 17, Jalan Kempas 4,	11, Wan Lee Road		
Taman Perindustrian Tengeh Indah,	Singapore 627943		
81200 Johor Bahru, Johor, Malaysia.			